

Filter Summary Report: CG,TIA,Automated,simple,Z1,Z5

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Contents

1 Examined $H(z)$ for CG TIA Automated simple Z1 Z5: $\frac{Z_1 Z_5 g_m - Z_1}{2Z_1 g_m + 1}$

$$H(z) = \frac{Z_1 Z_5 g_m - Z_1}{2Z_1 g_m + 1}$$

2 AP

3 BP

3.1 BP-1 $Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \infty, \infty, \infty, R_5, \infty \right)$

$$H(s) = \frac{s (L_1 R_5 g_m - L_1)}{C_1 L_1 s^2 + 2L_1 g_m s + 1}$$

Parameters:

Q: $\frac{\sqrt{C_1}}{2\sqrt{L_1} g_m}$
 wo: $\frac{1}{\sqrt{C_1} \sqrt{L_1}}$
 bandwidth: $\frac{2g_m}{C_1}$
 K-LP: 0
 K-HP: 0
 K-BP: $\frac{R_5 g_m - 1}{2g_m}$
 Qz: None
 Wz: None

3.2 BP-2 $Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \infty, \infty, \infty, R_5, \infty \right)$

$$H(s) = \frac{s (L_1 R_1 R_5 g_m - L_1 R_1)}{C_1 L_1 R_1 s^2 + R_1 + s (2L_1 R_1 g_m + L_1)}$$

Parameters:

Q: $\frac{\sqrt{C_1} R_1}{2\sqrt{L_1} R_1 g_m + \sqrt{L_1}}$
 wo: $\frac{1}{\sqrt{C_1} \sqrt{L_1}}$
 bandwidth: $\frac{2\sqrt{L_1} R_1 g_m + \sqrt{L_1}}{C_1 \sqrt{L_1} R_1}$
 K-LP: 0
 K-HP: 0
 K-BP: $\frac{R_1 R_5 g_m - R_1}{2R_1 g_m + 1}$
 Qz: None
 Wz: None

4 BS

4.1 BS-1 $Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \infty, \infty, \infty, R_5, \infty \right)$

$$H(s) = \frac{R_5 g_m + s^2 (C_1 L_1 R_5 g_m - C_1 L_1) - 1}{2C_1 L_1 g_m s^2 + C_1 s + 2g_m}$$

Parameters:

Q: $\frac{2\sqrt{L_1} g_m}{\sqrt{C_1}}$
 wo: $\frac{1}{\sqrt{C_1} \sqrt{L_1}}$
 bandwidth: $\frac{1}{2L_1 g_m}$
 K-LP: $\frac{R_5 g_m - 1}{2g_m}$
 K-HP: $\frac{R_5 g_m - 1}{2g_m}$

K-BP: 0
Qz: None
Wz: $\frac{1}{\sqrt{C_1}\sqrt{L_1}}$

$$4.2 \quad \text{BS-2} \quad Z(s) = \left(\frac{R_1(C_1 L_1 s^2 + 1)}{C_1 L_1 s^2 + C_1 R_1 s + 1}, \infty, \infty, \infty, R_5, \infty \right)$$

$$H(s) = \frac{R_1 R_5 g_m - R_1 + s^2 (C_1 L_1 R_1 R_5 g_m - C_1 L_1 R_1)}{C_1 R_1 s + 2 R_1 g_m + s^2 (2 C_1 L_1 R_1 g_m + C_1 L_1) + 1}$$

Parameters:

Q: $\frac{2\sqrt{L_1}R_1g_m+\sqrt{L_1}}{\sqrt{C_1}R_1}$
wo: $\frac{1}{\sqrt{C_1}\sqrt{L_1}}$
bandwidth: $\frac{R_1}{\sqrt{L_1}(2\sqrt{L_1}R_1g_m+\sqrt{L_1})}$
K-LP: $\frac{R_1R_5g_m-R_1}{2R_1g_m+1}$
K-HP: $\frac{R_1R_5g_m-R_1}{2R_1g_m+1}$
K-BP: 0
Qz: None
Wz: $\frac{1}{\sqrt{C_1}\sqrt{L_1}}$

5 GE

$$5.1 \quad \text{GE-1} \quad Z(s) = \left(R_1, \infty, \infty, \infty, \frac{L_5 R_5 s}{C_5 L_5 R_5 s^2 + L_5 s + R_5}, \infty \right)$$

$$H(s) = \frac{-C_5 L_5 R_1 R_5 s^2 - R_1 R_5 + s (L_5 R_1 R_5 g_m - L_5 R_1)}{2 R_1 R_5 g_m + R_5 + s^2 (2 C_5 L_5 R_1 R_5 g_m + C_5 L_5 R_5) + s (2 L_5 R_1 g_m + L_5)}$$

Parameters:

Q: $\frac{\sqrt{C_5}R_5}{\sqrt{L_5}}$
wo: $\frac{1}{\sqrt{C_5}\sqrt{L_5}}$
bandwidth: $\frac{1}{C_5 R_5}$
K-LP: $-\frac{R_1}{2R_1g_m+1}$
K-HP: $-\frac{R_1}{2R_1g_m+1}$
K-BP: $\frac{R_1R_5g_m-R_1}{2R_1g_m+1}$
Qz: $-\frac{\sqrt{C_5}R_5}{\sqrt{L_5}R_5g_m-\sqrt{L_5}}$
Wz: $\frac{1}{\sqrt{C_5}\sqrt{L_5}}$

$$5.2 \quad \text{GE-2} \quad Z(s) = \left(R_1, \infty, \infty, \infty, \frac{R_5(C_5 L_5 s^2 + 1)}{C_5 L_5 s^2 + C_5 R_5 s + 1}, \infty \right)$$

$$H(s) = \frac{-C_5 R_1 R_5 s + R_1 R_5 g_m - R_1 + s^2 (C_5 L_5 R_1 R_5 g_m - C_5 L_5 R_1)}{2 R_1 g_m + s^2 (2 C_5 L_5 R_1 g_m + C_5 L_5) + s (2 C_5 R_1 R_5 g_m + C_5 R_5) + 1}$$

Parameters:

Q: $\frac{\sqrt{L_5}}{\sqrt{C_5}R_5}$
wo: $\frac{1}{\sqrt{C_5}\sqrt{L_5}}$
bandwidth: $\frac{R_5}{L_5}$
K-LP: $\frac{R_1R_5g_m-R_1}{2R_1g_m+1}$
K-HP: $\frac{R_1R_5g_m-R_1}{2R_1g_m+1}$
K-BP: $-\frac{R_1}{2R_1g_m+1}$
Qz: $\frac{-\sqrt{L_5}R_5g_m+\sqrt{L_5}}{\sqrt{C_5}R_5}$
Wz: $\frac{1}{\sqrt{C_5}\sqrt{L_5}}$

5.3 GE-3 $Z(s) = \left(L_1 s + R_1 + \frac{1}{C_1 s}, \infty, \infty, \infty, R_5, \infty \right)$

Parameters:

$$\begin{aligned} \text{Q: } & \frac{2\sqrt{L_1 g_m}}{2\sqrt{C_1 R_1 g_m} + \sqrt{C_1}} \\ \text{wo: } & \frac{1}{\sqrt{C_1} \sqrt{L_1}} \\ \text{bandwidth: } & \frac{2\sqrt{C_1 R_1 g_m} + \sqrt{C_1}}{2\sqrt{C_1} L_1 g_m} \\ \text{K-LP: } & \frac{R_5 g_m - 1}{2g_m} \\ \text{K-HP: } & \frac{R_5 g_m - 1}{2g_m} \\ \text{K-BP: } & \frac{R_1 R_5 g_m - R_1}{2R_1 g_m + 1} \\ \text{Qz: } & \frac{\sqrt{L_1}}{\sqrt{C_1} R_1} \\ \text{Wz: } & \frac{1}{\sqrt{C_1} \sqrt{L_1}} \end{aligned}$$

5.4 GE-4 $Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1}, \infty, \infty, \infty, R_5, \infty \right)$

Parameters:

$$\begin{aligned} \text{Q: } & \frac{2\sqrt{C_1 R_1 g_m} + \sqrt{C_1}}{2\sqrt{L_1 g_m}} \\ \text{wo: } & \frac{1}{\sqrt{C_1} \sqrt{L_1}} \\ \text{bandwidth: } & \frac{2g_m}{\sqrt{C_1} (2\sqrt{C_1 R_1 g_m} + \sqrt{C_1})} \\ \text{K-LP: } & \frac{R_1 R_5 g_m - R_1}{2R_1 g_m + 1} \\ \text{K-HP: } & \frac{R_1 R_5 g_m - R_1}{2R_1 g_m + 1} \\ \text{K-BP: } & \frac{R_5 g_m - 1}{2g_m} \\ \text{Qz: } & \frac{\sqrt{C_1} R_1}{\sqrt{L_1}} \\ \text{Wz: } & \frac{1}{\sqrt{C_1} \sqrt{L_1}} \end{aligned}$$

6 HP

7 LP

8 X-INVALID-NUMER

8.1 X-INVALID-NUMER-1 $Z(s) = \left(L_1 s, \infty, \infty, \infty, \frac{R_5}{C_5 R_5 s + 1}, \infty \right)$

Parameters:

$$\begin{aligned} \text{Q: } & \frac{\sqrt{2}\sqrt{C_5}\sqrt{L_1}\sqrt{R_5 g_m}\sqrt{\frac{1}{g_m}}}{C_5 R_5 + 2L_1 g_m} \\ \text{wo: } & \frac{\sqrt{2}\sqrt{\frac{1}{g_m}}}{2\sqrt{C_5}\sqrt{L_1}\sqrt{R_5}} \\ \text{bandwidth: } & \frac{C_5 R_5 + 2L_1 g_m}{2C_5 L_1 R_5 g_m} \\ \text{K-LP: } & 0 \\ \text{K-HP: } & -\frac{1}{2g_m} \\ \text{K-BP: } & \frac{L_1 R_5 g_m - L_1}{C_5 R_5 + 2L_1 g_m} \end{aligned}$$

$$H(s) = \frac{R_5 g_m + s^2 (C_1 L_1 R_5 g_m - C_1 L_1) + s (C_1 R_1 R_5 g_m - C_1 R_1) - 1}{2C_1 L_1 g_m s^2 + 2g_m + s (2C_1 R_1 g_m + C_1)}$$

$$H(s) = \frac{R_1 R_5 g_m - R_1 + s^2 (C_1 L_1 R_1 R_5 g_m - C_1 L_1 R_1) + s (L_1 R_5 g_m - L_1)}{2L_1 g_m s + 2R_1 g_m + s^2 (2C_1 L_1 R_1 g_m + C_1 L_1) + 1}$$

$$H(s) = \frac{-C_5 L_1 R_5 s^2 + s (L_1 R_5 g_m - L_1)}{2C_5 L_1 R_5 g_m s^2 + s (C_5 R_5 + 2L_1 g_m) + 1}$$

Qz: None
Wz: None

8.2 X-INVALID-NUMER-2 $Z(s) = \left(\frac{1}{C_1 s}, \infty, \infty, \infty, \frac{R_5}{C_5 R_5 s + 1}, \infty \right)$

$$H(s) = \frac{-C_5 R_5 s + R_5 g_m - 1}{C_1 C_5 R_5 s^2 + 2g_m + s(C_1 + 2C_5 R_5 g_m)}$$

Parameters:

Q: $\frac{\sqrt{2}\sqrt{C_1}\sqrt{C_5}\sqrt{R_5}\sqrt{g_m}}{C_1 + 2C_5 R_5 g_m}$
 wo: $\frac{\sqrt{2}\sqrt{g_m}}{\sqrt{C_1}\sqrt{C_5}\sqrt{R_5}}$
 bandwidth: $\frac{C_1 + 2C_5 R_5 g_m}{C_1 C_5 R_5}$
 K-LP: $\frac{R_5 g_m - 1}{2g_m}$
 K-HP: 0
 K-BP: $-\frac{C_5 R_5}{C_1 + 2C_5 R_5 g_m}$
 Qz: None
 Wz: None

8.3 X-INVALID-NUMER-3 $Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \infty, \infty, \frac{R_5}{C_5 R_5 s + 1}, \infty \right)$

$$H(s) = \frac{-C_5 R_1 R_5 s + R_1 R_5 g_m - R_1}{C_1 C_5 R_1 R_5 s^2 + 2R_1 g_m + s(C_1 R_1 + 2C_5 R_1 R_5 g_m + C_5 R_5) + 1}$$

Parameters:

Q: $\frac{\sqrt{C_1}\sqrt{C_5}\sqrt{R_1}\sqrt{R_5}\sqrt{2R_1 g_m + 1}}{C_1 R_1 + 2C_5 R_1 R_5 g_m + C_5 R_5}$
 wo: $\frac{\sqrt{2R_1 g_m + 1}}{\sqrt{C_1}\sqrt{C_5}\sqrt{R_1}\sqrt{R_5}}$
 bandwidth: $\frac{C_1 R_1 + 2C_5 R_1 R_5 g_m + C_5 R_5}{C_1 C_5 R_1 R_5}$
 K-LP: $\frac{R_1 R_5 g_m - R_1}{2R_1 g_m + 1}$
 K-HP: 0
 K-BP: $-\frac{C_5 R_1 R_5}{C_1 R_1 + 2C_5 R_1 R_5 g_m + C_5 R_5}$
 Qz: None
 Wz: None

8.4 X-INVALID-NUMER-4 $Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \infty, \infty, \infty, \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{-C_5 L_1 s + L_1 g_m}{C_1 C_5 L_1 s^2 + 2C_5 L_1 g_m s + C_5}$$

Parameters:

Q: $\frac{\sqrt{C_1}}{2\sqrt{L_1}g_m}$
 wo: $\frac{1}{\sqrt{C_1}\sqrt{L_1}}$
 bandwidth: $\frac{2g_m}{C_1}$
 K-LP: $\frac{L_1 g_m}{C_5}$
 K-HP: 0
 K-BP: $-\frac{1}{2g_m}$
 Qz: None
 Wz: None

8.5 X-INVALID-NUMER-5 $Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \infty, \infty, \infty, R_5 + \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{L_1 g_m + s (C_5 L_1 R_5 g_m - C_5 L_1)}{C_1 C_5 L_1 s^2 + 2 C_5 L_1 g_m s + C_5}$$

Parameters:

Q: $\frac{\sqrt{C_1}}{2\sqrt{L_1} g_m}$
 wo: $\frac{1}{\sqrt{C_1} \sqrt{L_1}}$
 bandwidth: $\frac{2g_m}{C_1}$
 K-LP: $\frac{L_1 g_m}{C_5}$
 K-HP: 0
 K-BP: $\frac{R_5 g_m - 1}{2g_m}$
 Qz: None
 Wz: None

8.6 X-INVALID-NUMER-6 $Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \infty, \infty, \infty, \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{-C_5 L_1 R_1 s + L_1 R_1 g_m}{C_1 C_5 L_1 R_1 s^2 + C_5 R_1 + s (2 C_5 L_1 R_1 g_m + C_5 L_1)}$$

Parameters:

Q: $\frac{\sqrt{C_1} R_1}{2\sqrt{L_1} R_1 g_m + \sqrt{L_1}}$
 wo: $\frac{1}{\sqrt{C_1} \sqrt{L_1}}$
 bandwidth: $\frac{2\sqrt{L_1} R_1 g_m + \sqrt{L_1}}{C_1 \sqrt{L_1} R_1}$
 K-LP: $\frac{L_1 g_m}{C_5}$
 K-HP: 0
 K-BP: $-\frac{R_1}{2R_1 g_m + 1}$
 Qz: None
 Wz: None

8.7 X-INVALID-NUMER-7 $Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \infty, \infty, \infty, R_5 + \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{L_1 R_1 g_m + s (C_5 L_1 R_1 R_5 g_m - C_5 L_1 R_1)}{C_1 C_5 L_1 R_1 s^2 + C_5 R_1 + s (2 C_5 L_1 R_1 g_m + C_5 L_1)}$$

Parameters:

Q: $\frac{\sqrt{C_1} R_1}{2\sqrt{L_1} R_1 g_m + \sqrt{L_1}}$
 wo: $\frac{1}{\sqrt{C_1} \sqrt{L_1}}$
 bandwidth: $\frac{2\sqrt{L_1} R_1 g_m + \sqrt{L_1}}{C_1 \sqrt{L_1} R_1}$
 K-LP: $\frac{L_1 g_m}{C_5}$
 K-HP: 0
 K-BP: $\frac{R_1 R_5 g_m - R_1}{2R_1 g_m + 1}$
 Qz: None
 Wz: None

9 X-INVALID-ORDER

9.1 X-INVALID-ORDER-1 $Z(s) = (R_1, \infty, \infty, \infty, R_5, \infty)$

$$H(s) = \frac{R_1 R_5 g_m - R_1}{2R_1 g_m + 1}$$

$$\mathbf{9.2 \quad X-INVALID-ORDER-2} \quad Z(s) = \left(R_1, \infty, \infty, \infty, \frac{1}{C_5 s}, \infty \right)$$

$$H(s) = \frac{-C_5 R_1 s + R_1 g_m}{s(2C_5 R_1 g_m + C_5)}$$

$$\mathbf{9.3 \quad X-INVALID-ORDER-3} \quad Z(s) = \left(R_1, \infty, \infty, \infty, \frac{R_5}{C_5 R_5 s + 1}, \infty \right)$$

$$H(s) = \frac{-C_5 R_1 R_5 s + R_1 R_5 g_m - R_1}{2R_1 g_m + s(2C_5 R_1 R_5 g_m + C_5 R_5) + 1}$$

$$\mathbf{9.4 \quad X-INVALID-ORDER-4} \quad Z(s) = \left(R_1, \infty, \infty, \infty, R_5 + \frac{1}{C_5 s}, \infty \right)$$

$$H(s) = \frac{R_1 g_m + s(C_5 R_1 R_5 g_m - C_5 R_1)}{s(2C_5 R_1 g_m + C_5)}$$

$$\mathbf{9.5 \quad X-INVALID-ORDER-5} \quad Z(s) = \left(R_1, \infty, \infty, \infty, L_5 s + \frac{1}{C_5 s}, \infty \right)$$

$$H(s) = \frac{C_5 L_5 R_1 g_m s^2 - C_5 R_1 s + R_1 g_m}{s(2C_5 R_1 g_m + C_5)}$$

$$\mathbf{9.6 \quad X-INVALID-ORDER-6} \quad Z(s) = \left(R_1, \infty, \infty, \infty, \frac{L_5 s}{C_5 L_5 s^2 + 1}, \infty \right)$$

$$H(s) = \frac{-C_5 L_5 R_1 s^2 + L_5 R_1 g_m s - R_1}{2R_1 g_m + s^2(2C_5 L_5 R_1 g_m + C_5 L_5) + 1}$$

$$\mathbf{9.7 \quad X-INVALID-ORDER-7} \quad Z(s) = \left(R_1, \infty, \infty, \infty, L_5 s + R_5 + \frac{1}{C_5 s}, \infty \right)$$

$$H(s) = \frac{C_5 L_5 R_1 g_m s^2 + R_1 g_m + s(C_5 R_1 R_5 g_m - C_5 R_1)}{s(2C_5 R_1 g_m + C_5)}$$

$$\mathbf{9.8 \quad X-INVALID-ORDER-8} \quad Z(s) = \left(R_1, \infty, \infty, \infty, \frac{C_5 L_5 R_5 s^2 + L_5 s + R_5}{C_5 L_5 s^2 + 1}, \infty \right)$$

$$H(s) = \frac{L_5 R_1 g_m s + R_1 R_5 g_m - R_1 + s^2(C_5 L_5 R_1 R_5 g_m - C_5 L_5 R_1)}{2R_1 g_m + s^2(2C_5 L_5 R_1 g_m + C_5 L_5) + 1}$$

$$\mathbf{9.9 \quad X-INVALID-ORDER-9} \quad Z(s) = (L_1 s, \infty, \infty, \infty, R_5, \infty)$$

$$H(s) = \frac{s(L_1 R_5 g_m - L_1)}{2L_1 g_m s + 1}$$

$$\mathbf{9.10 \quad X-INVALID-ORDER-10} \quad Z(s) = \left(L_1 s, \infty, \infty, \infty, \frac{1}{C_5 s}, \infty \right)$$

$$H(s) = \frac{-C_5 L_1 s + L_1 g_m}{2C_5 L_1 g_m s + C_5}$$

$$\mathbf{9.11 \quad X-INVALID-ORDER-11} \quad Z(s) = \left(L_1 s, \infty, \infty, \infty, R_5 + \frac{1}{C_5 s}, \infty \right)$$

$$H(s) = \frac{L_1 g_m + s(C_5 L_1 R_5 g_m - C_5 L_1)}{2C_5 L_1 g_m s + C_5}$$

$$\mathbf{9.12 \quad X-INVALID-ORDER-12} \quad Z(s) = \left(L_1 s, \quad \infty, \quad \infty, \quad \infty, \quad L_5 s + \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{C_5 L_1 L_5 g_m s^2 - C_5 L_1 s + L_1 g_m}{2 C_5 L_1 g_m s + C_5}$$

$$\mathbf{9.13 \quad X-INVALID-ORDER-13} \quad Z(s) = \left(L_1 s, \quad \infty, \quad \infty, \quad \infty, \quad \frac{L_5 s}{C_5 L_5 s^2 + 1}, \quad \infty \right)$$

$$H(s) = \frac{-C_5 L_1 L_5 s^3 + L_1 L_5 g_m s^2 - L_1 s}{2 C_5 L_1 L_5 g_m s^3 + C_5 L_5 s^2 + 2 L_1 g_m s + 1}$$

$$\mathbf{9.14 \quad X-INVALID-ORDER-14} \quad Z(s) = \left(L_1 s, \quad \infty, \quad \infty, \quad \infty, \quad L_5 s + R_5 + \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{C_5 L_1 L_5 g_m s^2 + L_1 g_m + s (C_5 L_1 R_5 g_m - C_5 L_1)}{2 C_5 L_1 g_m s + C_5}$$

$$\mathbf{9.15 \quad X-INVALID-ORDER-15} \quad Z(s) = \left(L_1 s, \quad \infty, \quad \infty, \quad \infty, \quad \frac{L_5 R_5 s}{C_5 L_5 R_5 s^2 + L_5 s + R_5}, \quad \infty \right)$$

$$H(s) = \frac{-C_5 L_1 L_5 R_5 s^3 - L_1 R_5 s + s^2 (L_1 L_5 R_5 g_m - L_1 L_5)}{2 C_5 L_1 L_5 R_5 g_m s^3 + R_5 + s^2 (C_5 L_5 R_5 + 2 L_1 L_5 g_m) + s (2 L_1 R_5 g_m + L_5)}$$

$$\mathbf{9.16 \quad X-INVALID-ORDER-16} \quad Z(s) = \left(L_1 s, \quad \infty, \quad \infty, \quad \infty, \quad \frac{C_5 L_5 R_5 s^2 + L_5 s + R_5}{C_5 L_5 s^2 + 1}, \quad \infty \right)$$

$$H(s) = \frac{L_1 L_5 g_m s^2 + s^3 (C_5 L_1 L_5 R_5 g_m - C_5 L_1 L_5) + s (L_1 R_5 g_m - L_1)}{2 C_5 L_1 L_5 g_m s^3 + C_5 L_5 s^2 + 2 L_1 g_m s + 1}$$

$$\mathbf{9.17 \quad X-INVALID-ORDER-17} \quad Z(s) = \left(L_1 s, \quad \infty, \quad \infty, \quad \infty, \quad \frac{R_5 (C_5 L_5 s^2 + 1)}{C_5 L_5 s^2 + C_5 R_5 s + 1}, \quad \infty \right)$$

$$H(s) = \frac{-C_5 L_1 R_5 s^2 + s^3 (C_5 L_1 L_5 R_5 g_m - C_5 L_1 L_5) + s (L_1 R_5 g_m - L_1)}{2 C_5 L_1 L_5 g_m s^3 + s^2 (2 C_5 L_1 R_5 g_m + C_5 L_5) + s (C_5 R_5 + 2 L_1 g_m) + 1}$$

$$\mathbf{9.18 \quad X-INVALID-ORDER-18} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \infty, \quad R_5, \quad \infty \right)$$

$$H(s) = \frac{R_5 g_m - 1}{C_1 s + 2 g_m}$$

$$\mathbf{9.19 \quad X-INVALID-ORDER-19} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \infty, \quad \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{-C_5 s + g_m}{C_1 C_5 s^2 + 2 C_5 g_m s}$$

$$\mathbf{9.20 \quad X-INVALID-ORDER-20} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \infty, \quad R_5 + \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{g_m + s (C_5 R_5 g_m - C_5)}{C_1 C_5 s^2 + 2 C_5 g_m s}$$

$$\mathbf{9.21 \quad X-INVALID-ORDER-21} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \infty, \quad L_5 s + \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{C_5 L_5 g_m s^2 - C_5 s + g_m}{C_1 C_5 s^2 + 2 C_5 g_m s}$$

$$\mathbf{9.22 \quad X-INVALID-ORDER-22} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \infty, \quad \frac{L_5 s}{C_5 L_5 s^2 + 1}, \quad \infty \right)$$

$$H(s) = \frac{-C_5 L_5 s^2 + L_5 g_m s - 1}{C_1 C_5 L_5 s^3 + C_1 s + 2C_5 L_5 g_m s^2 + 2g_m}$$

$$\mathbf{9.23 \quad X-INVALID-ORDER-23} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \infty, \quad L_5 s + R_5 + \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{C_5 L_5 g_m s^2 + g_m + s (C_5 R_5 g_m - C_5)}{C_1 C_5 s^2 + 2C_5 g_m s}$$

$$\mathbf{9.24 \quad X-INVALID-ORDER-24} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \infty, \quad \frac{L_5 R_5 s}{C_5 L_5 R_5 s^2 + L_5 s + R_5}, \quad \infty \right)$$

$$H(s) = \frac{-C_5 L_5 R_5 s^2 - R_5 + s (L_5 R_5 g_m - L_5)}{C_1 C_5 L_5 R_5 s^3 + 2R_5 g_m + s^2 (C_1 L_5 + 2C_5 L_5 R_5 g_m) + s (C_1 R_5 + 2L_5 g_m)}$$

$$\mathbf{9.25 \quad X-INVALID-ORDER-25} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \infty, \quad \frac{C_5 L_5 R_5 s^2 + L_5 s + R_5}{C_5 L_5 s^2 + 1}, \quad \infty \right)$$

$$H(s) = \frac{L_5 g_m s + R_5 g_m + s^2 (C_5 L_5 R_5 g_m - C_5 L_5) - 1}{C_1 C_5 L_5 s^3 + C_1 s + 2C_5 L_5 g_m s^2 + 2g_m}$$

$$\mathbf{9.26 \quad X-INVALID-ORDER-26} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \infty, \quad \frac{R_5 (C_5 L_5 s^2 + 1)}{C_5 L_5 s^2 + C_5 R_5 s + 1}, \quad \infty \right)$$

$$H(s) = \frac{-C_5 R_5 s + R_5 g_m + s^2 (C_5 L_5 R_5 g_m - C_5 L_5) - 1}{C_1 C_5 L_5 s^3 + 2g_m + s^2 (C_1 C_5 R_5 + 2C_5 L_5 g_m) + s (C_1 + 2C_5 R_5 g_m)}$$

$$\mathbf{9.27 \quad X-INVALID-ORDER-27} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \quad \infty, \quad \infty, \quad \infty, \quad R_5, \quad \infty \right)$$

$$H(s) = \frac{R_1 R_5 g_m - R_1}{C_1 R_1 s + 2R_1 g_m + 1}$$

$$\mathbf{9.28 \quad X-INVALID-ORDER-28} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \quad \infty, \quad \infty, \quad \infty, \quad \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{-C_5 R_1 s + R_1 g_m}{C_1 C_5 R_1 s^2 + s (2C_5 R_1 g_m + C_5)}$$

$$\mathbf{9.29 \quad X-INVALID-ORDER-29} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \quad \infty, \quad \infty, \quad \infty, \quad R_5 + \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{R_1 g_m + s (C_5 R_1 R_5 g_m - C_5 R_1)}{C_1 C_5 R_1 s^2 + s (2C_5 R_1 g_m + C_5)}$$

$$\mathbf{9.30 \quad X-INVALID-ORDER-30} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \quad \infty, \quad \infty, \quad \infty, \quad L_5 s + \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{C_5 L_5 R_1 g_m s^2 - C_5 R_1 s + R_1 g_m}{C_1 C_5 R_1 s^2 + s (2C_5 R_1 g_m + C_5)}$$

$$\mathbf{9.31 \quad X-INVALID-ORDER-31} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \quad \infty, \quad \infty, \quad \infty, \quad \frac{L_5 s}{C_5 L_5 s^2 + 1}, \quad \infty \right)$$

$$H(s) = \frac{-C_5 L_5 R_1 s^2 + L_5 R_1 g_m s - R_1}{C_1 C_5 L_5 R_1 s^3 + C_1 R_1 s + 2R_1 g_m + s^2 (2C_5 L_5 R_1 g_m + C_5 L_5) + 1}$$

$$\mathbf{9.32 \quad X-INVALID-ORDER-32} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \infty, \infty, L_5 s + R_5 + \frac{1}{C_5 s}, \infty \right)$$

$$H(s) = \frac{C_5 L_5 R_1 g_m s^2 + R_1 g_m + s (C_5 R_1 R_5 g_m - C_5 R_1)}{C_1 C_5 R_1 s^2 + s (2 C_5 R_1 g_m + C_5)}$$

$$\mathbf{9.33 \quad X-INVALID-ORDER-33} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \infty, \infty, \frac{L_5 R_5 s}{C_5 L_5 R_5 s^2 + L_5 s + R_5}, \infty \right)$$

$$H(s) = \frac{-C_5 L_5 R_1 R_5 s^2 - R_1 R_5 + s (L_5 R_1 R_5 g_m - L_5 R_1)}{C_1 C_5 L_5 R_1 R_5 s^3 + 2 R_1 R_5 g_m + R_5 + s^2 (C_1 L_5 R_1 + 2 C_5 L_5 R_1 R_5 g_m + C_5 L_5 R_5) + s (C_1 R_1 R_5 + 2 L_5 R_1 g_m + L_5)}$$

$$\mathbf{9.34 \quad X-INVALID-ORDER-34} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \infty, \infty, \frac{C_5 L_5 R_5 s^2 + L_5 s + R_5}{C_5 L_5 s^2 + 1}, \infty \right)$$

$$H(s) = \frac{L_5 R_1 g_m s + R_1 R_5 g_m - R_1 + s^2 (C_5 L_5 R_1 R_5 g_m - C_5 L_5 R_1)}{C_1 C_5 L_5 R_1 s^3 + C_1 R_1 s + 2 R_1 g_m + s^2 (2 C_5 L_5 R_1 g_m + C_5 L_5) + 1}$$

$$\mathbf{9.35 \quad X-INVALID-ORDER-35} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \infty, \infty, \frac{R_5 (C_5 L_5 s^2 + 1)}{C_5 L_5 s^2 + C_5 R_5 s + 1}, \infty \right)$$

$$H(s) = \frac{-C_5 R_1 R_5 s + R_1 R_5 g_m - R_1 + s^2 (C_5 L_5 R_1 R_5 g_m - C_5 L_5 R_1)}{C_1 C_5 L_5 R_1 s^3 + 2 R_1 g_m + s^2 (C_1 C_5 R_1 R_5 + 2 C_5 L_5 R_1 g_m + C_5 L_5) + s (C_1 R_1 + 2 C_5 R_1 R_5 g_m + C_5 R_5) + 1}$$

$$\mathbf{9.36 \quad X-INVALID-ORDER-36} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \infty, \infty, \infty, R_5, \infty \right)$$

$$H(s) = \frac{R_5 g_m + s (C_1 R_1 R_5 g_m - C_1 R_1) - 1}{2 g_m + s (2 C_1 R_1 g_m + C_1)}$$

$$\mathbf{9.37 \quad X-INVALID-ORDER-37} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \infty, \infty, \infty, \frac{1}{C_5 s}, \infty \right)$$

$$H(s) = \frac{-C_1 C_5 R_1 s^2 + g_m + s (C_1 R_1 g_m - C_5)}{2 C_5 g_m s + s^2 (2 C_1 C_5 R_1 g_m + C_1 C_5)}$$

$$\mathbf{9.38 \quad X-INVALID-ORDER-38} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \infty, \infty, \infty, R_5 + \frac{1}{C_5 s}, \infty \right)$$

$$H(s) = \frac{g_m + s^2 (C_1 C_5 R_1 R_5 g_m - C_1 C_5 R_1) + s (C_1 R_1 g_m + C_5 R_5 g_m - C_5)}{2 C_5 g_m s + s^2 (2 C_1 C_5 R_1 g_m + C_1 C_5)}$$

$$\mathbf{9.39 \quad X-INVALID-ORDER-39} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \infty, \infty, \infty, L_5 s + \frac{1}{C_5 s}, \infty \right)$$

$$H(s) = \frac{C_1 C_5 L_5 R_1 g_m s^3 + g_m + s^2 (-C_1 C_5 R_1 + C_5 L_5 g_m) + s (C_1 R_1 g_m - C_5)}{2 C_5 g_m s + s^2 (2 C_1 C_5 R_1 g_m + C_1 C_5)}$$

$$\mathbf{9.40 \quad X-INVALID-ORDER-40} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \infty, \infty, \infty, \frac{L_5 s}{C_5 L_5 s^2 + 1}, \infty \right)$$

$$H(s) = \frac{-C_1 C_5 L_5 R_1 s^3 + s^2 (C_1 L_5 R_1 g_m - C_5 L_5) + s (-C_1 R_1 + L_5 g_m) - 1}{2 C_5 L_5 g_m s^2 + 2 g_m + s^3 (2 C_1 C_5 L_5 R_1 g_m + C_1 C_5 L_5) + s (2 C_1 R_1 g_m + C_1)}$$

$$9.41 \quad \text{X-INVALID-ORDER-41} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \infty, \quad L_5 s + R_5 + \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_5 L_5 R_1 g_m s^3 + g_m + s^2 (C_1 C_5 R_1 R_5 g_m - C_1 C_5 R_1 + C_5 L_5 g_m) + s (C_1 R_1 g_m + C_5 R_5 g_m - C_5)}{2 C_5 g_m s + s^2 (2 C_1 C_5 R_1 g_m + C_1 C_5)}$$

$$9.42 \quad \text{X-INVALID-ORDER-42} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \infty, \quad \frac{L_5 R_5 s}{C_5 L_5 R_5 s^2 + L_5 s + R_5}, \quad \infty \right)$$

$$H(s) = \frac{-C_1 C_5 L_5 R_1 R_5 s^3 - R_5 + s^2 (C_1 L_5 R_1 R_5 g_m - C_1 L_5 R_1 - C_5 L_5 R_5) + s (-C_1 R_1 R_5 + L_5 R_5 g_m - L_5)}{2 R_5 g_m + s^3 (2 C_1 C_5 L_5 R_1 R_5 g_m + C_1 C_5 L_5 R_5) + s^2 (2 C_1 L_5 R_1 g_m + C_1 L_5 + 2 C_5 L_5 R_5 g_m) + s (2 C_1 R_1 R_5 g_m + C_1 R_5 + 2 L_5 g_m)}$$

$$9.43 \quad \text{X-INVALID-ORDER-43} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \infty, \quad \frac{C_5 L_5 R_5 s^2 + L_5 s + R_5}{C_5 L_5 s^2 + 1}, \quad \infty \right)$$

$$H(s) = \frac{R_5 g_m + s^3 (C_1 C_5 L_5 R_1 R_5 g_m - C_1 C_5 L_5 R_1) + s^2 (C_1 L_5 R_1 g_m + C_5 L_5 R_5 g_m - C_5 L_5) + s (C_1 R_1 R_5 g_m - C_1 R_1 + L_5 g_m) - 1}{2 C_5 L_5 g_m s^2 + 2 g_m + s^3 (2 C_1 C_5 L_5 R_1 g_m + C_1 C_5 L_5) + s (2 C_1 R_1 g_m + C_1)}$$

$$9.44 \quad \text{X-INVALID-ORDER-44} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \infty, \quad \frac{R_5 (C_5 L_5 s^2 + 1)}{C_5 L_5 s^2 + C_5 R_5 s + 1}, \quad \infty \right)$$

$$H(s) = \frac{R_5 g_m + s^3 (C_1 C_5 L_5 R_1 R_5 g_m - C_1 C_5 L_5 R_1) + s^2 (-C_1 C_5 R_1 R_5 + C_5 L_5 R_5 g_m - C_5 L_5) + s (C_1 R_1 R_5 g_m - C_1 R_1 - C_5 R_5) - 1}{2 g_m + s^3 (2 C_1 C_5 L_5 R_1 g_m + C_1 C_5 L_5) + s^2 (2 C_1 C_5 R_1 R_5 g_m + C_1 C_5 R_5 + 2 C_5 L_5 g_m) + s (2 C_1 R_1 g_m + C_1 + 2 C_5 R_5 g_m)}$$

$$9.45 \quad \text{X-INVALID-ORDER-45} \quad Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \infty, \quad \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{-C_1 C_5 L_1 s^3 + C_1 L_1 g_m s^2 - C_5 s + g_m}{2 C_1 C_5 L_1 g_m s^3 + C_1 C_5 s^2 + 2 C_5 g_m s}$$

$$9.46 \quad \text{X-INVALID-ORDER-46} \quad Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \infty, \quad \frac{R_5}{C_5 R_5 s + 1}, \quad \infty \right)$$

$$H(s) = \frac{-C_1 C_5 L_1 R_5 s^3 - C_5 R_5 s + R_5 g_m + s^2 (C_1 L_1 R_5 g_m - C_1 L_1) - 1}{2 C_1 C_5 L_1 R_5 g_m s^3 + 2 g_m + s^2 (C_1 C_5 R_5 + 2 C_1 L_1 g_m) + s (C_1 + 2 C_5 R_5 g_m)}$$

$$9.47 \quad \text{X-INVALID-ORDER-47} \quad Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \infty, \quad R_5 + \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{C_1 L_1 g_m s^2 + g_m + s^3 (C_1 C_5 L_1 R_5 g_m - C_1 C_5 L_1) + s (C_5 R_5 g_m - C_5)}{2 C_1 C_5 L_1 g_m s^3 + C_1 C_5 s^2 + 2 C_5 g_m s}$$

$$9.48 \quad \text{X-INVALID-ORDER-48} \quad Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \infty, \quad L_5 s + \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_5 L_1 L_5 g_m s^4 - C_1 C_5 L_1 s^3 - C_5 s + g_m + s^2 (C_1 L_1 g_m + C_5 L_5 g_m)}{2 C_1 C_5 L_1 g_m s^3 + C_1 C_5 s^2 + 2 C_5 g_m s}$$

$$9.49 \quad \text{X-INVALID-ORDER-49} \quad Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \infty, \quad \frac{L_5 s}{C_5 L_5 s^2 + 1}, \quad \infty \right)$$

$$H(s) = \frac{-C_1 C_5 L_1 L_5 s^4 + C_1 L_1 L_5 g_m s^3 + L_5 g_m s + s^2 (-C_1 L_1 - C_5 L_5) - 1}{2 C_1 C_5 L_1 L_5 g_m s^4 + C_1 C_5 L_5 s^3 + C_1 s + 2 g_m + s^2 (2 C_1 L_1 g_m + 2 C_5 L_5 g_m)}$$

$$\mathbf{9.50 \quad X-INVALID-ORDER-50} \quad Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \infty, \quad L_5 s + R_5 + \frac{1}{C_5 s}, \quad \infty \right)$$

$$H(s) = \frac{C_1 C_5 L_1 L_5 g_m s^4 + g_m + s^3 (C_1 C_5 L_1 R_5 g_m - C_1 C_5 L_1) + s^2 (C_1 L_1 g_m + C_5 L_5 g_m) + s (C_5 R_5 g_m - C_5)}{2 C_1 C_5 L_1 g_m s^3 + C_1 C_5 s^2 + 2 C_5 g_m s}$$

$$\mathbf{9.51 \quad X-INVALID-ORDER-51} \quad Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \infty, \quad \frac{L_5 R_5 s}{C_5 L_5 R_5 s^2 + L_5 s + R_5}, \quad \infty \right)$$

$$H(s) = \frac{-C_1 C_5 L_1 L_5 R_5 s^4 - R_5 + s^3 (C_1 L_1 L_5 R_5 g_m - C_1 L_1 L_5) + s^2 (-C_1 L_1 R_5 - C_5 L_5 R_5) + s (L_5 R_5 g_m - L_5)}{2 C_1 C_5 L_1 L_5 R_5 g_m s^4 + 2 R_5 g_m + s^3 (C_1 C_5 L_5 R_5 + 2 C_1 L_1 L_5 g_m) + s^2 (2 C_1 L_1 R_5 g_m + C_1 L_5 + 2 C_5 L_5 R_5 g_m) + s (C_1 R_5 + 2 L_5 g_m)}$$

$$\mathbf{9.52 \quad X-INVALID-ORDER-52} \quad Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \infty, \quad \frac{C_5 L_5 R_5 s^2 + L_5 s + R_5}{C_5 L_5 s^2 + 1}, \quad \infty \right)$$

$$H(s) = \frac{C_1 L_1 L_5 g_m s^3 + L_5 g_m s + R_5 g_m + s^4 (C_1 C_5 L_1 L_5 R_5 g_m - C_1 C_5 L_1 L_5) + s^2 (C_1 L_1 R_5 g_m - C_1 L_1 + C_5 L_5 R_5 g_m - C_5 L_5) - 1}{2 C_1 C_5 L_1 L_5 g_m s^4 + C_1 C_5 L_5 s^3 + C_1 s + 2 g_m + s^2 (2 C_1 L_1 g_m + 2 C_5 L_5 g_m)}$$

$$\mathbf{9.53 \quad X-INVALID-ORDER-53} \quad Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \infty, \quad \frac{R_5 (C_5 L_5 s^2 + 1)}{C_5 L_5 s^2 + C_5 R_5 s + 1}, \quad \infty \right)$$

$$H(s) = \frac{-C_1 C_5 L_1 R_5 s^3 - C_5 R_5 s + R_5 g_m + s^4 (C_1 C_5 L_1 L_5 R_5 g_m - C_1 C_5 L_1 L_5) + s^2 (C_1 L_1 R_5 g_m - C_1 L_1 + C_5 L_5 R_5 g_m - C_5 L_5) - 1}{2 C_1 C_5 L_1 L_5 g_m s^4 + 2 g_m + s^3 (2 C_1 C_5 L_1 R_5 g_m + C_1 C_5 L_5) + s^2 (C_1 C_5 R_5 + 2 C_1 L_1 g_m + 2 C_5 L_5 g_m) + s (C_1 + 2 C_5 R_5 g_m)}$$

$$\mathbf{9.54 \quad X-INVALID-ORDER-54} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \infty, \quad \infty, \quad \frac{R_5}{C_5 R_5 s + 1}, \quad \infty \right)$$

$$H(s) = \frac{-C_5 L_1 R_5 s^2 + s (L_1 R_5 g_m - L_1)}{C_1 C_5 L_1 R_5 s^3 + s^2 (C_1 L_1 + 2 C_5 L_1 R_5 g_m) + s (C_5 R_5 + 2 L_1 g_m) + 1}$$

$$\mathbf{9.55 \quad X-INVALID-ORDER-55} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \infty, \quad \infty, \quad \frac{L_5 s}{C_5 L_5 s^2 + 1}, \quad \infty \right)$$

$$H(s) = \frac{-C_5 L_1 L_5 s^3 + L_1 L_5 g_m s^2 - L_1 s}{C_1 C_5 L_1 L_5 s^4 + 2 C_5 L_1 L_5 g_m s^3 + 2 L_1 g_m s + s^2 (C_1 L_1 + C_5 L_5) + 1}$$

$$\mathbf{9.56 \quad X-INVALID-ORDER-56} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \infty, \quad \infty, \quad \frac{L_5 R_5 s}{C_5 L_5 R_5 s^2 + L_5 s + R_5}, \quad \infty \right)$$

$$H(s) = \frac{-C_5 L_1 L_5 R_5 s^3 - L_1 R_5 s + s^2 (L_1 L_5 R_5 g_m - L_1 L_5)}{C_1 C_5 L_1 L_5 R_5 s^4 + R_5 + s^3 (C_1 L_1 L_5 + 2 C_5 L_1 L_5 R_5 g_m) + s^2 (C_1 L_1 R_5 + C_5 L_5 R_5 + 2 L_1 L_5 g_m) + s (2 L_1 R_5 g_m + L_5)}$$

$$\mathbf{9.57 \quad X-INVALID-ORDER-57} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \infty, \quad \infty, \quad \frac{C_5 L_5 R_5 s^2 + L_5 s + R_5}{C_5 L_5 s^2 + 1}, \quad \infty \right)$$

$$H(s) = \frac{L_1 L_5 g_m s^2 + s^3 (C_5 L_1 L_5 R_5 g_m - C_5 L_1 L_5) + s (L_1 R_5 g_m - L_1)}{C_1 C_5 L_1 L_5 s^4 + 2 C_5 L_1 L_5 g_m s^3 + 2 L_1 g_m s + s^2 (C_1 L_1 + C_5 L_5) + 1}$$

$$\mathbf{9.58 \quad X-INVALID-ORDER-58} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \infty, \quad \infty, \quad \frac{R_5 (C_5 L_5 s^2 + 1)}{C_5 L_5 s^2 + C_5 R_5 s + 1}, \quad \infty \right)$$

$$H(s) = \frac{-C_5 L_1 R_5 s^2 + s^3 (C_5 L_1 L_5 R_5 g_m - C_5 L_1 L_5) + s (L_1 R_5 g_m - L_1)}{C_1 C_5 L_1 L_5 s^4 + s^3 (C_1 C_5 L_1 R_5 + 2 C_5 L_1 L_5 g_m) + s^2 (C_1 L_1 + 2 C_5 L_1 R_5 g_m + C_5 L_5) + s (C_5 R_5 + 2 L_1 g_m) + 1}$$

9.59 X-INVALID-ORDER-59 $Z(s) = \left(L_1s + R_1 + \frac{1}{C_1s}, \infty, \infty, \infty, \frac{1}{C_5s}, \infty \right)$

$$H(s) = \frac{-C_1C_5L_1s^3 + g_m + s^2(-C_1C_5R_1 + C_1L_1g_m) + s(C_1R_1g_m - C_5)}{2C_1C_5L_1g_ms^3 + 2C_5g_ms + s^2(2C_1C_5R_1g_m + C_1C_5)}$$

9.60 X-INVALID-ORDER-60 $Z(s) = \left(L_1s + R_1 + \frac{1}{C_1s}, \infty, \infty, \infty, \frac{R_5}{C_5R_5s+1}, \infty \right)$

$$H(s) = \frac{-C_1C_5L_1R_5s^3 + R_5g_m + s^2(-C_1C_5R_1R_5 + C_1L_1R_5g_m - C_1L_1) + s(C_1R_1R_5g_m - C_1R_1 - C_5R_5) - 1}{2C_1C_5L_1R_5g_ms^3 + 2g_m + s^2(2C_1C_5R_1R_5g_m + C_1C_5R_5 + 2C_1L_1g_m) + s(2C_1R_1g_m + C_1 + 2C_5R_5g_m)}$$

9.61 X-INVALID-ORDER-61 $Z(s) = \left(L_1s + R_1 + \frac{1}{C_1s}, \infty, \infty, \infty, R_5 + \frac{1}{C_5s}, \infty \right)$

$$H(s) = \frac{g_m + s^3(C_1C_5L_1R_5g_m - C_1C_5L_1) + s^2(C_1C_5R_1R_5g_m - C_1C_5R_1 + C_1L_1g_m) + s(C_1R_1g_m + C_5R_5g_m - C_5)}{2C_1C_5L_1g_ms^3 + 2C_5g_ms + s^2(2C_1C_5R_1g_m + C_1C_5)}$$

9.62 X-INVALID-ORDER-62 $Z(s) = \left(L_1s + R_1 + \frac{1}{C_1s}, \infty, \infty, \infty, L_5s + \frac{1}{C_5s}, \infty \right)$

$$H(s) = \frac{C_1C_5L_1L_5g_ms^4 + g_m + s^3(-C_1C_5L_1 + C_1C_5L_5R_1g_m) + s^2(-C_1C_5R_1 + C_1L_1g_m + C_5L_5g_m) + s(C_1R_1g_m - C_5)}{2C_1C_5L_1g_ms^3 + 2C_5g_ms + s^2(2C_1C_5R_1g_m + C_1C_5)}$$

9.63 X-INVALID-ORDER-63 $Z(s) = \left(L_1s + R_1 + \frac{1}{C_1s}, \infty, \infty, \infty, \frac{L_5s}{C_5L_5s^2+1}, \infty \right)$

$$H(s) = \frac{-C_1C_5L_1L_5s^4 + s^3(-C_1C_5L_5R_1 + C_1L_1L_5g_m) + s^2(-C_1L_1 + C_1L_5R_1g_m - C_5L_5) + s(-C_1R_1 + L_5g_m) - 1}{2C_1C_5L_1L_5g_ms^4 + 2g_m + s^3(2C_1C_5L_5R_1g_m + C_1C_5L_5) + s^2(2C_1L_1g_m + 2C_5L_5g_m) + s(2C_1R_1g_m + C_1)}$$

9.64 X-INVALID-ORDER-64 $Z(s) = \left(L_1s + R_1 + \frac{1}{C_1s}, \infty, \infty, \infty, L_5s + R_5 + \frac{1}{C_5s}, \infty \right)$

$$H(s) = \frac{C_1C_5L_1L_5g_ms^4 + g_m + s^3(C_1C_5L_1R_5g_m - C_1C_5L_1 + C_1C_5L_5R_1g_m) + s^2(C_1C_5R_1R_5g_m - C_1C_5R_1 + C_1L_1g_m + C_5L_5g_m) + s(C_1R_1g_m + C_5R_5g_m - C_5)}{2C_1C_5L_1g_ms^3 + 2C_5g_ms + s^2(2C_1C_5R_1g_m + C_1C_5)}$$

9.65 X-INVALID-ORDER-65 $Z(s) = \left(L_1s + R_1 + \frac{1}{C_1s}, \infty, \infty, \infty, \frac{L_5R_5s}{C_5L_5R_5s^2+L_5s+R_5}, \infty \right)$

$$H(s) = \frac{-C_1C_5L_1L_5R_5s^4 - R_5 + s^3(-C_1C_5L_5R_1R_5 + C_1L_1L_5R_5g_m - C_1L_1L_5) + s^2(-C_1L_1R_5 + C_1L_5R_1R_5g_m - C_1L_5R_1 - C_5L_5R_5) + s(-C_1R_1R_5 + L_5R_5g_m - L_5)}{2C_1C_5L_1L_5R_5g_ms^4 + 2R_5g_m + s^3(2C_1C_5L_5R_1R_5g_m + C_1C_5L_5R_5 + 2C_1L_1L_5g_m) + s^2(2C_1L_1R_5g_m + 2C_1L_5R_1g_m + C_1L_5 + 2C_5L_5R_5g_m) + s(2C_1R_1R_5g_m + C_1R_5 + 2L_5g_m)}$$

9.66 X-INVALID-ORDER-66 $Z(s) = \left(L_1s + R_1 + \frac{1}{C_1s}, \infty, \infty, \infty, \frac{C_5L_5R_5s^2+L_5s+R_5}{C_5L_5s^2+1}, \infty \right)$

$$H(s) = \frac{R_5g_m + s^4(C_1C_5L_1L_5R_5g_m - C_1C_5L_1L_5) + s^3(C_1C_5L_5R_1R_5g_m - C_1C_5L_5R_1 + C_1L_1L_5g_m) + s^2(C_1L_1R_5g_m - C_1L_1 + C_1L_5R_1g_m + C_5L_5R_5g_m - C_5L_5) + s(C_1R_1R_5g_m - C_1R_1 + L_5g_m) - 1}{2C_1C_5L_1L_5g_ms^4 + 2g_m + s^3(2C_1C_5L_5R_1g_m + C_1C_5L_5) + s^2(2C_1L_1g_m + 2C_5L_5g_m) + s(2C_1R_1g_m + C_1)}$$

9.67 X-INVALID-ORDER-67 $Z(s) = \left(L_1s + R_1 + \frac{1}{C_1s}, \infty, \infty, \infty, \frac{R_5(C_5L_5s^2+1)}{C_5L_5s^2+C_5R_5s+1}, \infty \right)$

$$H(s) = \frac{R_5g_m + s^4(C_1C_5L_1L_5R_5g_m - C_1C_5L_1L_5) + s^3(-C_1C_5L_1R_5 + C_1C_5L_5R_1R_5g_m - C_1C_5L_5R_1) + s^2(-C_1C_5R_1R_5 + C_1L_1R_5g_m - C_1L_1 + C_5L_5R_5g_m - C_5L_5) + s(C_1R_1R_5g_m - C_1R_1 - C_5R_5) - 1}{2C_1C_5L_1L_5g_ms^4 + 2g_m + s^3(2C_1C_5L_1R_5g_m + 2C_1C_5L_5R_1g_m + C_1C_5L_5) + s^2(2C_1C_5R_1R_5g_m + C_1C_5R_5 + 2C_1L_1g_m + 2C_5L_5g_m) + s(2C_1R_1g_m + C_1 + 2C_5R_5g_m)}$$

$$\mathbf{9.68 \quad X-INVALID-ORDER-68} \quad Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \infty, \infty, \infty, \frac{R_5}{C_5 R_5 s + 1}, \infty \right)$$

$$H(s) = \frac{-C_5 L_1 R_1 R_5 s^2 + s (L_1 R_1 R_5 g_m - L_1 R_1)}{C_1 C_5 L_1 R_1 R_5 s^3 + R_1 + s^2 (C_1 L_1 R_1 + 2C_5 L_1 R_1 R_5 g_m + C_5 L_1 R_5) + s (C_5 R_1 R_5 + 2L_1 R_1 g_m + L_1)}$$

$$\mathbf{9.69 \quad X-INVALID-ORDER-69} \quad Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \infty, \infty, \infty, \frac{L_5 s}{C_5 L_5 s^2 + 1}, \infty \right)$$

$$H(s) = \frac{-C_5 L_1 L_5 R_1 s^3 + L_1 L_5 R_1 g_m s^2 - L_1 R_1 s}{C_1 C_5 L_1 L_5 R_1 s^4 + R_1 + s^3 (2C_5 L_1 L_5 R_1 g_m + C_5 L_1 L_5) + s^2 (C_1 L_1 R_1 + C_5 L_5 R_1) + s (2L_1 R_1 g_m + L_1)}$$

$$\mathbf{9.70 \quad X-INVALID-ORDER-70} \quad Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \infty, \infty, \infty, \frac{L_5 R_5 s}{C_5 L_5 R_5 s^2 + L_5 s + R_5}, \infty \right)$$

$$H(s) = \frac{-C_5 L_1 L_5 R_1 R_5 s^3 - L_1 R_1 R_5 s + s^2 (L_1 L_5 R_1 R_5 g_m - L_1 L_5 R_1)}{C_1 C_5 L_1 L_5 R_1 R_5 s^4 + R_1 R_5 + s^3 (C_1 L_1 L_5 R_1 + 2C_5 L_1 L_5 R_1 R_5 g_m + C_5 L_1 L_5 R_5) + s^2 (C_1 L_1 R_1 R_5 + C_5 L_5 R_1 R_5 + 2L_1 L_5 R_1 g_m + L_1 L_5) + s (2L_1 R_1 R_5 g_m + L_1 R_5 + L_5 R_1)}$$

$$\mathbf{9.71 \quad X-INVALID-ORDER-71} \quad Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \infty, \infty, \infty, \frac{C_5 L_5 R_5 s^2 + L_5 s + R_5}{C_5 L_5 s^2 + 1}, \infty \right)$$

$$H(s) = \frac{L_1 L_5 R_1 g_m s^2 + s^3 (C_5 L_1 L_5 R_1 R_5 g_m - C_5 L_1 L_5 R_1) + s (L_1 R_1 R_5 g_m - L_1 R_1)}{C_1 C_5 L_1 L_5 R_1 s^4 + R_1 + s^3 (2C_5 L_1 L_5 R_1 g_m + C_5 L_1 L_5) + s^2 (C_1 L_1 R_1 + C_5 L_5 R_1) + s (2L_1 R_1 g_m + L_1)}$$

$$\mathbf{9.72 \quad X-INVALID-ORDER-72} \quad Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \infty, \infty, \infty, \frac{R_5 (C_5 L_5 s^2 + 1)}{C_5 L_5 s^2 + C_5 R_5 s + 1}, \infty \right)$$

$$H(s) = \frac{-C_5 L_1 R_1 R_5 s^2 + s^3 (C_5 L_1 L_5 R_1 R_5 g_m - C_5 L_1 L_5 R_1) + s (L_1 R_1 R_5 g_m - L_1 R_1)}{C_1 C_5 L_1 L_5 R_1 s^4 + R_1 + s^3 (C_1 C_5 L_1 R_1 R_5 + 2C_5 L_1 L_5 R_1 g_m + C_5 L_1 L_5) + s^2 (C_1 L_1 R_1 + 2C_5 L_1 R_1 R_5 g_m + C_5 L_1 R_5 + C_5 L_5 R_1) + s (C_5 R_1 R_5 + 2L_1 R_1 g_m + L_1)}$$

$$\mathbf{9.73 \quad X-INVALID-ORDER-73} \quad Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1}, \infty, \infty, \infty, \frac{1}{C_5 s}, \infty \right)$$

$$H(s) = \frac{-C_1 C_5 L_1 R_1 s^3 + R_1 g_m + s^2 (C_1 L_1 R_1 g_m - C_5 L_1) + s (-C_5 R_1 + L_1 g_m)}{2C_5 L_1 g_m s^2 + s^3 (2C_1 C_5 L_1 R_1 g_m + C_1 C_5 L_1) + s (2C_5 R_1 g_m + C_5)}$$

$$\mathbf{9.74 \quad X-INVALID-ORDER-74} \quad Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1}, \infty, \infty, \infty, \frac{R_5}{C_5 R_5 s + 1}, \infty \right)$$

$$H(s) = \frac{-C_1 C_5 L_1 R_1 R_5 s^3 + R_1 R_5 g_m - R_1 + s^2 (C_1 L_1 R_1 R_5 g_m - C_1 L_1 R_1 - C_5 L_1 R_5) + s (-C_5 R_1 R_5 + L_1 R_5 g_m - L_1)}{2R_1 g_m + s^3 (2C_1 C_5 L_1 R_1 R_5 g_m + C_1 C_5 L_1 R_5) + s^2 (2C_1 L_1 R_1 g_m + C_1 L_1 + 2C_5 L_1 R_5 g_m) + s (2C_5 R_1 R_5 g_m + C_5 R_5 + 2L_1 g_m) + 1}$$

$$\mathbf{9.75 \quad X-INVALID-ORDER-75} \quad Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1}, \infty, \infty, \infty, R_5 + \frac{1}{C_5 s}, \infty \right)$$

$$H(s) = \frac{R_1 g_m + s^3 (C_1 C_5 L_1 R_1 R_5 g_m - C_1 C_5 L_1 R_1) + s^2 (C_1 L_1 R_1 g_m + C_5 L_1 R_5 g_m - C_5 L_1) + s (C_5 R_1 R_5 g_m - C_5 R_1 + L_1 g_m)}{2C_5 L_1 g_m s^2 + s^3 (2C_1 C_5 L_1 R_1 g_m + C_1 C_5 L_1) + s (2C_5 R_1 g_m + C_5)}$$

$$\mathbf{9.76 \quad X-INVALID-ORDER-76} \quad Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1}, \infty, \infty, \infty, L_5 s + \frac{1}{C_5 s}, \infty \right)$$

$$H(s) = \frac{C_1 C_5 L_1 L_5 R_1 g_m s^4 + R_1 g_m + s^3 (-C_1 C_5 L_1 R_1 + C_5 L_1 L_5 g_m) + s^2 (C_1 L_1 R_1 g_m - C_5 L_1 + C_5 L_5 R_1 g_m) + s (-C_5 R_1 + L_1 g_m)}{2C_5 L_1 g_m s^2 + s^3 (2C_1 C_5 L_1 R_1 g_m + C_1 C_5 L_1) + s (2C_5 R_1 g_m + C_5)}$$

$$\mathbf{9.77 \quad X-INVALID-ORDER-77} \quad Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1}, \infty, \infty, \infty, \frac{L_5 s}{C_5 L_5 s^2 + 1}, \infty \right)$$

$$H(s) = \frac{-C_1 C_5 L_1 L_5 R_1 s^4 - R_1 + s^3 (C_1 L_1 L_5 R_1 g_m - C_5 L_1 L_5) + s^2 (-C_1 L_1 R_1 - C_5 L_5 R_1 + L_1 L_5 g_m) + s (-L_1 + L_5 R_1 g_m)}{2 C_5 L_1 L_5 g_m s^3 + 2 L_1 g_m s + 2 R_1 g_m + s^4 (2 C_1 C_5 L_1 L_5 R_1 g_m + C_1 C_5 L_1 L_5) + s^2 (2 C_1 L_1 R_1 g_m + C_1 L_1 + 2 C_5 L_5 R_1 g_m + C_5 L_5) + 1}$$

$$\mathbf{9.78 \quad X-INVALID-ORDER-78} \quad Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1}, \infty, \infty, \infty, L_5 s + R_5 + \frac{1}{C_5 s}, \infty \right)$$

$$H(s) = \frac{C_1 C_5 L_1 L_5 R_1 g_m s^4 + R_1 g_m + s^3 (C_1 C_5 L_1 R_1 R_5 g_m - C_1 C_5 L_1 R_1 + C_5 L_1 L_5 g_m) + s^2 (C_1 L_1 R_1 g_m + C_5 L_1 R_5 g_m - C_5 L_1 + C_5 L_5 R_1 g_m) + s (C_5 R_1 R_5 g_m - C_5 R_1 + L_1 g_m)}{2 C_5 L_1 g_m s^2 + s^3 (2 C_1 C_5 L_1 R_1 g_m + C_1 C_5 L_1) + s (2 C_5 R_1 g_m + C_5)}$$

$$\mathbf{9.79 \quad X-INVALID-ORDER-79} \quad Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1}, \infty, \infty, \infty, \frac{L_5 R_5 s}{C_5 L_5 R_5 s^2 + L_5 s + R_5}, \infty \right)$$

$$H(s) = \frac{-C_1 C_5 L_1 L_5 R_1 R_5 s^4 - R_1 R_5 + s^3 (C_1 L_1 L_5 R_1 R_5 g_m - C_1 L_1 L_5 R_1 - C_5 L_1 L_5 R_5) + s^2 (-C_1 L_1 R_1 R_5 - C_5 L_5 R_1 R_5 + L_1 L_5 R_5 g_m - L_1 L_5) + s (-L_1 R_5 + L_5 R_1 R_5 g_m - L_5 R_1)}{2 R_1 R_5 g_m + R_5 + s^4 (2 C_1 C_5 L_1 L_5 R_1 R_5 g_m + C_1 C_5 L_1 L_5 R_5) + s^3 (2 C_1 L_1 L_5 R_1 g_m + C_1 L_1 L_5 + 2 C_5 L_1 L_5 R_5 g_m) + s^2 (2 C_1 L_1 R_1 R_5 g_m + C_1 L_1 R_5 + 2 C_5 L_5 R_1 R_5 g_m + C_5 L_5 R_5 + 2 L_1 L_5 g_m) + s (2 L_1 R_5 g_m + 2 L_5 R_1 g_m + L_5)}$$

$$\mathbf{9.80 \quad X-INVALID-ORDER-80} \quad Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1}, \infty, \infty, \infty, \frac{C_5 L_5 R_5 s^2 + L_5 s + R_5}{C_5 L_5 s^2 + 1}, \infty \right)$$

$$H(s) = \frac{R_1 R_5 g_m - R_1 + s^4 (C_1 C_5 L_1 L_5 R_1 R_5 g_m - C_1 C_5 L_1 L_5 R_1) + s^3 (C_1 L_1 L_5 R_1 g_m + C_5 L_1 L_5 R_5 g_m - C_5 L_1 L_5) + s^2 (C_1 L_1 R_1 R_5 g_m - C_1 L_1 R_1 + C_5 L_5 R_1 R_5 g_m - C_5 L_5 R_1 + L_1 L_5 g_m) + s (L_1 R_5 g_m - L_1 + L_5 R_1 g_m)}{2 C_5 L_1 L_5 g_m s^3 + 2 L_1 g_m s + 2 R_1 g_m + s^4 (2 C_1 C_5 L_1 L_5 R_1 g_m + C_1 C_5 L_1 L_5) + s^2 (2 C_1 L_1 R_1 g_m + C_1 L_1 + 2 C_5 L_5 R_1 g_m + C_5 L_5) + 1}$$

$$\mathbf{9.81 \quad X-INVALID-ORDER-81} \quad Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1}, \infty, \infty, \infty, \frac{R_5 (C_5 L_5 s^2 + 1)}{C_5 L_5 s^2 + C_5 R_5 s + 1}, \infty \right)$$

$$H(s) = \frac{R_1 R_5 g_m - R_1 + s^4 (C_1 C_5 L_1 L_5 R_1 R_5 g_m - C_1 C_5 L_1 L_5 R_1) + s^3 (-C_1 C_5 L_1 R_1 R_5 + C_5 L_1 L_5 R_5 g_m - C_5 L_1 L_5) + s^2 (C_1 L_1 R_1 R_5 g_m - C_1 L_1 R_1 - C_5 L_1 R_5 + C_5 L_5 R_1 R_5 g_m - C_5 L_5 R_1) + s (-C_5 R_1 R_5 + L_1 R_5 g_m - L_1)}{2 R_1 g_m + s^4 (2 C_1 C_5 L_1 L_5 R_1 g_m + C_1 C_5 L_1 L_5) + s^3 (2 C_1 C_5 L_1 R_1 R_5 g_m + C_1 C_5 L_1 R_5 + 2 C_5 L_1 L_5 g_m) + s^2 (2 C_1 L_1 R_1 g_m + C_1 L_1 + 2 C_5 L_1 R_5 g_m + 2 C_5 L_5 R_1 g_m + C_5 L_5) + s (2 C_5 R_1 R_5 g_m + C_5 R_5 + 2 L_1 g_m) + 1}$$

$$\mathbf{9.82 \quad X-INVALID-ORDER-82} \quad Z(s) = \left(\frac{R_1 (C_1 L_1 s^2 + 1)}{C_1 L_1 s^2 + C_1 R_1 s + 1}, \infty, \infty, \infty, \frac{1}{C_5 s}, \infty \right)$$

$$H(s) = \frac{-C_1 C_5 L_1 R_1 s^3 + C_1 L_1 R_1 g_m s^2 - C_5 R_1 s + R_1 g_m}{C_1 C_5 R_1 s^2 + s^3 (2 C_1 C_5 L_1 R_1 g_m + C_1 C_5 L_1) + s (2 C_5 R_1 g_m + C_5)}$$

$$\mathbf{9.83 \quad X-INVALID-ORDER-83} \quad Z(s) = \left(\frac{R_1 (C_1 L_1 s^2 + 1)}{C_1 L_1 s^2 + C_1 R_1 s + 1}, \infty, \infty, \infty, \frac{R_5}{C_5 R_5 s + 1}, \infty \right)$$

$$H(s) = \frac{-C_1 C_5 L_1 R_1 R_5 s^3 - C_5 R_1 R_5 s + R_1 R_5 g_m - R_1 + s^2 (C_1 L_1 R_1 R_5 g_m - C_1 L_1 R_1)}{2 R_1 g_m + s^3 (2 C_1 C_5 L_1 R_1 R_5 g_m + C_1 C_5 L_1 R_5) + s^2 (C_1 C_5 R_1 R_5 + 2 C_1 L_1 R_1 g_m + C_1 L_1) + s (C_1 R_1 + 2 C_5 R_1 R_5 g_m + C_5 R_5) + 1}$$

$$\mathbf{9.84 \quad X-INVALID-ORDER-84} \quad Z(s) = \left(\frac{R_1 (C_1 L_1 s^2 + 1)}{C_1 L_1 s^2 + C_1 R_1 s + 1}, \infty, \infty, \infty, R_5 + \frac{1}{C_5 s}, \infty \right)$$

$$H(s) = \frac{C_1 L_1 R_1 g_m s^2 + R_1 g_m + s^3 (C_1 C_5 L_1 R_1 R_5 g_m - C_1 C_5 L_1 R_1) + s (C_5 R_1 R_5 g_m - C_5 R_1)}{C_1 C_5 R_1 s^2 + s^3 (2 C_1 C_5 L_1 R_1 g_m + C_1 C_5 L_1) + s (2 C_5 R_1 g_m + C_5)}$$

$$\mathbf{9.85 \quad X-INVALID-ORDER-85} \quad Z(s) = \left(\frac{R_1 (C_1 L_1 s^2 + 1)}{C_1 L_1 s^2 + C_1 R_1 s + 1}, \infty, \infty, \infty, L_5 s + \frac{1}{C_5 s}, \infty \right)$$

$$H(s) = \frac{C_1 C_5 L_1 L_5 R_1 g_m s^4 - C_1 C_5 L_1 R_1 s^3 - C_5 R_1 s + R_1 g_m + s^2 (C_1 L_1 R_1 g_m + C_5 L_5 R_1 g_m)}{C_1 C_5 R_1 s^2 + s^3 (2 C_1 C_5 L_1 R_1 g_m + C_1 C_5 L_1) + s (2 C_5 R_1 g_m + C_5)}$$

9.86 X-INVALID-ORDER-86 $Z(s) = \left(\frac{R_1(C_1 L_1 s^2 + 1)}{C_1 L_1 s^2 + C_1 R_1 s + 1}, \infty, \infty, \infty, \frac{L_5 s}{C_5 L_5 s^2 + 1}, \infty \right)$

$$H(s) = \frac{-C_1 C_5 L_1 L_5 R_1 s^4 + C_1 L_1 L_5 R_1 g_m s^3 + L_5 R_1 g_m s - R_1 + s^2 (-C_1 L_1 R_1 - C_5 L_5 R_1)}{C_1 C_5 L_5 R_1 s^3 + C_1 R_1 s + 2R_1 g_m + s^4 (2C_1 C_5 L_1 L_5 R_1 g_m + C_1 C_5 L_1 L_5) + s^2 (2C_1 L_1 R_1 g_m + C_1 L_1 + 2C_5 L_5 R_1 g_m + C_5 L_5) + 1}$$

9.87 X-INVALID-ORDER-87 $Z(s) = \left(\frac{R_1(C_1 L_1 s^2 + 1)}{C_1 L_1 s^2 + C_1 R_1 s + 1}, \infty, \infty, \infty, L_5 s + R_5 + \frac{1}{C_5 s}, \infty \right)$

$$H(s) = \frac{C_1 C_5 L_1 L_5 R_1 g_m s^4 + R_1 g_m + s^3 (C_1 C_5 L_1 R_1 R_5 g_m - C_1 C_5 L_1 R_1) + s^2 (C_1 L_1 R_1 g_m + C_5 L_5 R_1 g_m) + s (C_5 R_1 R_5 g_m - C_5 R_1)}{C_1 C_5 R_1 s^2 + s^3 (2C_1 C_5 L_1 R_1 g_m + C_1 C_5 L_1) + s (2C_5 R_1 g_m + C_5)}$$

9.88 X-INVALID-ORDER-88 $Z(s) = \left(\frac{R_1(C_1 L_1 s^2 + 1)}{C_1 L_1 s^2 + C_1 R_1 s + 1}, \infty, \infty, \infty, \frac{L_5 R_5 s}{C_5 L_5 R_5 s^2 + L_5 s + R_5}, \infty \right)$

$$H(s) = \frac{-C_1 C_5 L_1 L_5 R_1 R_5 s^4 - R_1 R_5 + s^3 (C_1 L_1 L_5 R_1 R_5 g_m - C_1 L_1 L_5 R_1) + s^2 (-C_1 L_1 R_1 R_5 - C_5 L_5 R_1 R_5) + s (L_5 R_1 R_5 g_m - L_5 R_1)}{2R_1 R_5 g_m + R_5 + s^4 (2C_1 C_5 L_1 L_5 R_1 R_5 g_m + C_1 C_5 L_1 L_5 R_5) + s^3 (C_1 C_5 L_5 R_1 R_5 + 2C_1 L_1 L_5 R_1 g_m + C_1 L_1 L_5) + s^2 (2C_1 L_1 R_1 R_5 g_m + C_1 L_1 R_5 + C_1 L_5 R_1 + 2C_5 L_5 R_1 R_5 g_m + C_5 L_5 R_5) + s (C_1 R_1 R_5 + 2L_5 R_1 g_m + L_5)}$$

9.89 X-INVALID-ORDER-89 $Z(s) = \left(\frac{R_1(C_1 L_1 s^2 + 1)}{C_1 L_1 s^2 + C_1 R_1 s + 1}, \infty, \infty, \infty, \frac{C_5 L_5 R_5 s^2 + L_5 s + R_5}{C_5 L_5 s^2 + 1}, \infty \right)$

$$H(s) = \frac{C_1 L_1 L_5 R_1 g_m s^3 + L_5 R_1 g_m s + R_1 R_5 g_m - R_1 + s^4 (C_1 C_5 L_1 L_5 R_1 R_5 g_m - C_1 C_5 L_1 L_5 R_1) + s^2 (C_1 L_1 R_1 R_5 g_m - C_1 L_1 R_1 + C_5 L_5 R_1 R_5 g_m - C_5 L_5 R_1)}{C_1 C_5 L_5 R_1 s^3 + C_1 R_1 s + 2R_1 g_m + s^4 (2C_1 C_5 L_1 L_5 R_1 g_m + C_1 C_5 L_1 L_5) + s^2 (2C_1 L_1 R_1 g_m + C_1 L_1 + 2C_5 L_5 R_1 g_m + C_5 L_5) + 1}$$

9.90 X-INVALID-ORDER-90 $Z(s) = \left(\frac{R_1(C_1 L_1 s^2 + 1)}{C_1 L_1 s^2 + C_1 R_1 s + 1}, \infty, \infty, \infty, \frac{R_5(C_5 L_5 s^2 + 1)}{C_5 L_5 s^2 + C_5 R_5 s + 1}, \infty \right)$

$$H(s) = \frac{-C_1 C_5 L_1 R_1 R_5 s^3 - C_5 R_1 R_5 s + R_1 R_5 g_m - R_1 + s^4 (C_1 C_5 L_1 L_5 R_1 R_5 g_m - C_1 C_5 L_1 L_5 R_1) + s^2 (C_1 L_1 R_1 R_5 g_m - C_1 L_1 R_1 + C_5 L_5 R_1 R_5 g_m - C_5 L_5 R_1)}{2R_1 g_m + s^4 (2C_1 C_5 L_1 L_5 R_1 g_m + C_1 C_5 L_1 L_5) + s^3 (2C_1 C_5 L_1 R_1 R_5 g_m + C_1 C_5 L_1 R_5 + C_1 C_5 L_5 R_1) + s^2 (C_1 C_5 R_1 R_5 + 2C_1 L_1 R_1 g_m + C_1 L_1 + 2C_5 L_5 R_1 g_m + C_5 L_5) + s (C_1 R_1 + 2C_5 R_1 R_5 g_m + C_5 R_5) + 1}$$

10 X-INVALID-WZ

10.1 X-INVALID-WZ-1 $Z(s) = \left(R_1 + \frac{1}{C_1 s}, \infty, \infty, \infty, \frac{R_5}{C_5 R_5 s + 1}, \infty \right)$

$$H(s) = \frac{-C_1 C_5 R_1 R_5 s^2 + R_5 g_m + s (C_1 R_1 R_5 g_m - C_1 R_1 - C_5 R_5) - 1}{2g_m + s^2 (2C_1 C_5 R_1 R_5 g_m + C_1 C_5 R_5) + s (2C_1 R_1 g_m + C_1 + 2C_5 R_5 g_m)}$$

Parameters:

Q: $\frac{2\sqrt{2}\sqrt{C_1}\sqrt{C_5}R_1\sqrt{R_5g_m}\sqrt{\frac{g_m}{2R_1g_m+1}}+\sqrt{2}\sqrt{C_1}\sqrt{C_5}\sqrt{R_5}\sqrt{\frac{g_m}{2R_1g_m+1}}}{2C_1R_1g_m+C_1+2C_5R_5g_m}$

wo: $\sqrt{2}\sqrt{\frac{g_m}{2C_1C_5R_1R_5g_m+C_1C_5R_5}}$

bandwidth: $\frac{\sqrt{2}\sqrt{\frac{g_m}{2C_1C_5R_1R_5g_m+C_1C_5R_5}}(2C_1R_1g_m+C_1+2C_5R_5g_m)}{2\sqrt{2}\sqrt{C_1}\sqrt{C_5}R_1\sqrt{R_5g_m}\sqrt{\frac{g_m}{2R_1g_m+1}}+\sqrt{2}\sqrt{C_1}\sqrt{C_5}\sqrt{R_5}\sqrt{\frac{g_m}{2R_1g_m+1}}}$

K-LP: $\frac{R_5g_m-1}{2g_m}$

K-HP: $-\frac{R_1}{2R_1g_m+1}$

K-BP: $\frac{C_1R_1R_5g_m-C_1R_1-C_5R_5}{2C_1R_1g_m+C_1+2C_5R_5g_m}$

Qz: None

Wz: $\frac{\sqrt{-R_5g_m+1}}{\sqrt{C_1}\sqrt{C_5}\sqrt{R_1}\sqrt{R_5}}$

10.2 X-INVALID-WZ-2 $Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \infty, \infty, \infty, L_5 s + \frac{1}{C_5 s}, \infty \right)$

Parameters:

$$\begin{aligned} \text{Q: } & \frac{\sqrt{C_1}}{2\sqrt{L_1}g_m} \\ \text{wo: } & \frac{1}{\sqrt{C_1}\sqrt{L_1}} \\ \text{bandwidth: } & \frac{2g_m}{C_1} \\ \text{K-LP: } & \frac{L_1 g_m}{C_5} \\ \text{K-HP: } & \frac{L_5 g_m}{C_1} \\ \text{K-BP: } & -\frac{1}{2g_m} \\ \text{Qz: } & \text{None} \\ \text{Wz: } & \frac{1}{\sqrt{C_5}\sqrt{L_5}} \end{aligned}$$

$$H(s) = \frac{C_5 L_1 L_5 g_m s^2 - C_5 L_1 s + L_1 g_m}{C_1 C_5 L_1 s^2 + 2C_5 L_1 g_m s + C_5}$$

10.3 X-INVALID-WZ-3 $Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \infty, \infty, \infty, L_5 s + R_5 + \frac{1}{C_5 s}, \infty \right)$

Parameters:

$$\begin{aligned} \text{Q: } & \frac{\sqrt{C_1}}{2\sqrt{L_1}g_m} \\ \text{wo: } & \frac{1}{\sqrt{C_1}\sqrt{L_1}} \\ \text{bandwidth: } & \frac{2g_m}{C_1} \\ \text{K-LP: } & \frac{L_1 g_m}{C_5} \\ \text{K-HP: } & \frac{L_5 g_m}{C_1} \\ \text{K-BP: } & \frac{R_5 g_m - 1}{2g_m} \\ \text{Qz: } & \text{None} \\ \text{Wz: } & \frac{1}{\sqrt{C_5}\sqrt{L_5}} \end{aligned}$$

$$H(s) = \frac{C_5 L_1 L_5 g_m s^2 + L_1 g_m + s(C_5 L_1 R_5 g_m - C_5 L_1)}{C_1 C_5 L_1 s^2 + 2C_5 L_1 g_m s + C_5}$$

10.4 X-INVALID-WZ-4 $Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \infty, \infty, \infty, L_5 s + \frac{1}{C_5 s}, \infty \right)$

Parameters:

$$\begin{aligned} \text{Q: } & \frac{\sqrt{C_1} R_1}{2\sqrt{L_1} R_1 g_m + \sqrt{L_1}} \\ \text{wo: } & \frac{1}{\sqrt{C_1}\sqrt{L_1}} \\ \text{bandwidth: } & \frac{2\sqrt{L_1} R_1 g_m + \sqrt{L_1}}{C_1 \sqrt{L_1} R_1} \\ \text{K-LP: } & \frac{L_1 g_m}{C_5} \\ \text{K-HP: } & \frac{L_5 g_m}{C_1} \\ \text{K-BP: } & -\frac{R_1}{2R_1 g_m + 1} \\ \text{Qz: } & \text{None} \\ \text{Wz: } & \frac{1}{\sqrt{C_5}\sqrt{L_5}} \end{aligned}$$

$$H(s) = \frac{C_5 L_1 L_5 R_1 g_m s^2 - C_5 L_1 R_1 s + L_1 R_1 g_m}{C_1 C_5 L_1 R_1 s^2 + C_5 R_1 + s(2C_5 L_1 R_1 g_m + C_5 L_1)}$$

10.5 X-INVALID-WZ-5 $Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \infty, \infty, \infty, L_5 s + R_5 + \frac{1}{C_5 s}, \infty \right)$

Parameters:

$$\begin{aligned} \text{Q: } & \frac{\sqrt{C_1} R_1}{2\sqrt{L_1} R_1 g_m + \sqrt{L_1}} \\ \text{wo: } & \frac{1}{\sqrt{C_1}\sqrt{L_1}} \end{aligned}$$

$$H(s) = \frac{C_5 L_1 L_5 R_1 g_m s^2 + L_1 R_1 g_m + s(C_5 L_1 R_1 R_5 g_m - C_5 L_1 R_1)}{C_1 C_5 L_1 R_1 s^2 + C_5 R_1 + s(2C_5 L_1 R_1 g_m + C_5 L_1)}$$

bandwidth: $\frac{2\sqrt{L_1R_1g_m}+\sqrt{L_1}}{C_1\sqrt{L_1R_1}}$

K-LP: $\frac{L_1g_m}{C_5}$

K-HP: $\frac{L_5g_m}{C_1}$

K-BP: $\frac{R_1R_5g_m-R_1}{2R_1g_m+1}$

Qz: None

Wz: $\frac{1}{\sqrt{C_5}\sqrt{L_5}}$

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